

Immersive Virtual Reality Training : Three Cases from the Danish Industry

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ABSTRACT

Effective industrial training has always been vital. Recently, the need for robust, safe, repeatable and cost-effective digital applications drove many industries to explore immersive technologies and especially Immersive Virtual Reality (IVR) as a possible solution. In this paper, we depict three such cases, from the Danish industry and showcase the motivation of such adoption, technological and design characteristics, alongside perception of its applicability.

Index Terms: I.3.7 [Computer Graphics]: Three-Dimensional Graphics and Realism—Virtual Reality;

1 INTRODUCTION

On average, organizations spent 10% of their budget on learning tools and technologies with the most frequent purchases on e-learning and systems, learning management systems, and simulations [10]. These investments in training activities allow organizations to adapt, compete, excel, innovate, produce, be safe, improve services, and reach goals [13]. Training is effectively deployed to decrease errors in such high-risk work settings as emergency rooms, aviation, and the military. Now, a new innovative wave of Immersive Technologies (Virtual/Augmented/Mixed Reality - VR/AR/MR) is being utilized as training tools. The need for such implementations reflects the requirements for cost-effective, safe, scalable, modular and, mobile systems. Yet, this is a new phenomenon, with unexplored issues such as the motivation for adoption, trends in design and technologies, assessment techniques, and perception of stakeholders.

- RQ1: What motivates industries to adopt IVR training?
- RQ2: What technological parameters and design elements are incorporated in the IVR training?
- RQ3: How is IVR training assessed and perceived by the stakeholders?

With this section being the introduction, the rest of the paper is structured as follows. Section 2 provides a short introduction to the theoretical background, Section 3 depicts the methodology, Section 4 the case organizations and the IVR applications, Section 5 the analysis and discussion, and Section 6 the conclusions.

2 THEORETICAL BACKGROUND

Effective industrial training has always been paramount. Literature, reports that the cost for fail-to-recall procedural tasks in a production environment is high, and errors can be catastrophic for the product and the overall production cost [9]. Similar shortcomings apply

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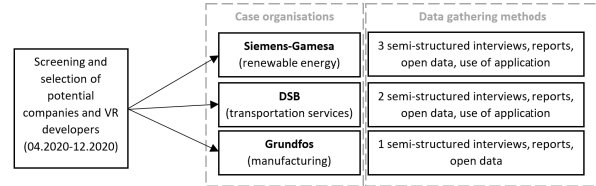


Figure 1: Case organizations and data gathering methods.

when errors occur beyond the production line, for example, assembly of machinery or installation of electronic components in the field of business or hard to approach locations, like offshore wind turbines or oil drilling rigs [23]. However, training is not as intuitive as it may seem. Literature portrays variations in the way one can design, deliver, and implement a training program [24]. Thus, it comes as no surprise to observe a great interest and investment from industry and academia on new innovative approaches to enhance training procedures. To this a new set of immersive technologies as Virtual Reality (VR); a fully digital environment, Augmented Reality (AR); an environment where the digital content augments the real world and Mixed Reality (MR); the combination of real and digital with interactive elements, has emerged to provide compelling solutions. Immersive technologies have been successfully utilized for example in health and safety [11, 12], disaster preparedness and response training [15, 22], surgery training [14, 25] and assembly tasks [8, 18]. Currently, Immersive Technologies are in a state of continuous development, with new affordable head-mounted displays for VR and AR glasses, advanced untethered communication capabilities, and increasing maturity in software development tools [7] and assessment techniques [19]. These technologies if utilized correctly can facilitate a risk-reduced, innovative, robust, cost-effective, repeatable practice environment [17, 20, 21]. The Danish industry sector has welcomed this opportunity, and in recent years a business ecosystem of VR developing companies (mainly SMEs) and big industrial partners (mainly in the renewable and services sectors) has been created. In this setting, the investigation described herein took place.

3 METHODOLOGY

The focus in Denmark was placed mainly due to the suitability to explore this phenomenon (availability of IVR cases) and the constrained travel due to the COVID19 restrictions that placed a barrier to explore cases in other countries. The three cases mentioned herein are the shortlist of many inspiring examples of IVR training we came across but these were chosen both for their representability of the IVR cases but also for the availability of case data.

In order to answer the three aforementioned research questions a thorough inspection of each case was performed including the gathering and analysis of a) semi-structured interviews, b) inspection of the IVR application and field studies, and c) open published data and available reports, as seen in Fig. 1.

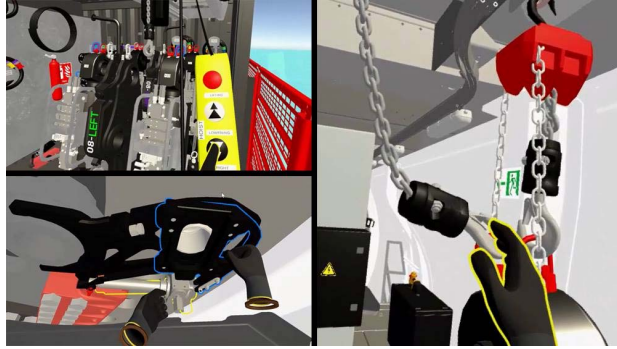


Figure 2: IVR training solution in Siemens-Gamesa. Retrieved from [5]

4 CASE ORGANISATIONS

4.1 Siemens-Gamesa

Siemens-Gamesa is a Spanish-German wind engineering company, which manufactures wind turbines and provides onshore and offshore wind services. Siemens-Gamesa is a key player and innovative pioneer in the renewable energy sector; they have installed products and technology in more than 90 countries. It is the world's second-largest wind turbine manufacturer.

Challenge Part of their offshore wind turbine installations is to train workers into performing specialised installations (referred to as "turning tool" training). Reported challenges of such training are the weather conditions and health and safety regulation when the training takes place in the offshore turbine in the sea and cost (renting the turning tool and transporting trainees and trainers) when the training is conducted in a simulated environment in the land.

Motivation One of the main objectives for the Siemens-Gamesa training department to adopt IVR training was to provide an easy to transport, safe, and repeatable training solution that will reduce the cost of training for technicians involved in the installation of wind turbines. The development of the IVR training was carried out by Kanda, a Danish VR development company [2] and was implemented in an iterative proof-of-concept process where funds were allocated as the solution matured.

Skills The skills that are taught with the IVR solution primarily involve procedural skills and familiarisation with the work environment and tools. This includes process training steps on how to assemble parts of the wind turbine, how to identify areas where objects like screws are placed and more complex skills like operating a telescopic crane among others. In the case of the telescopic crane, technicians need to learn the use of an internal container telescopic crane to lower down equipment from the container on the top of the wind turbine nacelle.

Design Elements Gamification and multi-user interactivity are integral parts of this IVR solution and has been designed to allow people to make mistakes. Time is not measured in any manner. Additionally, visual aids are present in the IVR training environment to guide the learner. For example, tools - objects that needed to be utilised to complete tasks were highlighted, as well as the areas where they were to be placed. The environment is not photorealistic but feedback from trainees and trainers suggest that the virtual environment is highly realistic, as seen in Fig. 2.

Technological Elements

- **Virtual Reality** – The HTC Vive is used for the currently deployed VR training. For most parts of the training, the interactions involve two hands (using the VR controllers). The

workspace is room-scale, as the trainees have to kneel down, walk around and turn. For further movement, they teleport around the VR environment.

- **Haptics** - Haptic feedback is used in the simulator to give vibrations when trainees walk into virtual walls. Still, there are no vibrations while interacting with objects. The interviewees however indicated that conveying weight and texture information during interaction with virtual objects would be helpful for skill training. Physical controllers were not used in order to reduce costs.
- **User Adaptation** – An interesting observation from the training sessions revealed that trainees with experience playing video games adapted to the training faster and seemed more enthusiastic in learning the VR interaction features and the use of the controller. On the other hand, there was not any significant difference between the aforementioned users and the rest of the trained workers in terms of learning performance. Nevertheless, the company expressed interest in adapting the IVR training aligning dissimilar trainee profiles.

Assessments There are no assessments or any kind of score-tracking features incorporated with the IVR experience. Instead, they rely on the trainer's observations to analyze and decide on whether the trainees are learning or not, mainly based on the progress observed. The trainers view the trainee's activity and progress through a desktop monitor.

Outcome The interviewees reported on the positive outcome of the application of IVR as a training tool and highlighted that has been rolled out in several training settings. However, as no metrics, comparison or statistics are available it was not clear if the IVR training has been more effective than traditional methods, but a documented benefit was that the time for renting out the turning tool equipment had been reduced from two weeks to one. Additionally, and mainly through the perceived notion of it beneficial use the investment and the IVR as a training technology was seen with a positive eye and considered as worth exploring further.

4.2 DSB

DSB is the largest Danish train operating company and the largest in Scandinavia. As of 2005, DSB employs about 9,000 people with an increasing number of services each year [1].

Challenge The company employs various roles to service, in the train, during a journey, amongst them the train operators. Part of the train operators' job is to manage, in a timely manner, opening and closing of doors, when the trains are stationary at stations and or in the case of an emergency. Training takes place by decommissioning the train and providing access to the trainer and trainees to proceed with the training scenarios.

Motivation The aforementioned method of training causes unwanted downtime in functioning trains and does not always provide the realistic scenario i.e. that of a functioning, noisy station is resembling real life. Thus, an IVR experience that may simulate such a scenario was created. The key objective is to train DSB train operators to learn the use of the whistle and door operation.

Design Elements The trainees learn the procedures related to managing the opening and closing of train doors in a linear single-user IVR experience, (no decision making or fine motor skills are taught). The virtual environment is a detailed representation of the train station platform, the train doors, and the immediate interior of the train coach behind the doors. The trainees navigate the virtual environment through teleportation while standing in one physical location. Visual cues, as seen in Fig. 3 are present to aid the trainee by displaying time and highlight points-of-interest. The trainees first undertake a theory class on the subject matter before the IVR

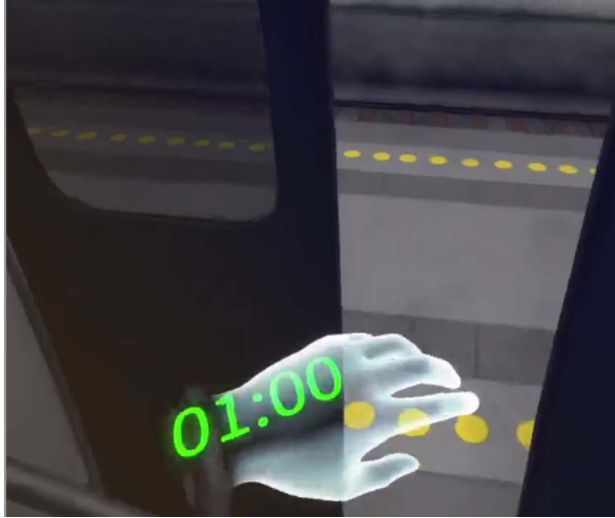


Figure 3: IVR training solution in DSB. Retrieved from [16]

experience. Training is delivered at the headquarters in batches of 12 trainees, who are split into groups of two. Each group is assigned an IVR headset and a tablet. There are two expert trainers who manage the training experience and aid the trainees reflect on what they learned.

Technological Elements

- *Virtual Reality* – The trainee uses an Oculus Quest and an iPad, where one trainee experiences IVR training in the Quest while the other observes the progression, and provides feedback.
- *Haptics* – Haptic feedback is present as an indicator of the trainee that certain actions were performed. For example, when a trainee opens the train doors gets vibration feedback to indicate successful completion. This is meant more like a nudge than to represent a physically realistic interaction.
- *User Adaptation* – No adaptation for specific users was identified.

Assessments There are no formal assessments for the IVR training. Completion time is logged and provided in the trainees, as well as the oral peer-feedback they get during the training but neither is analysed or compared with other relative data.

Outcome The interviewees perceived the IVR as a beneficial training method, still requiring development but due to the ever-increasing demand for fast training at a lower cost a step towards the right direction.

4.3 Grundfos

Grundfos is the largest pump manufacturer in the world, with more than 18,000 employees globally. The annual production of more than 16 million pump units, circulator pumps, submersible pumps, and centrifugal pumps [6].

Challenge Training employees in machine operation and safety procedures during the manufacturing process is a crucial task that benefits error and accident reduction. For such training machines are diverted from production to an idle state, increasing downtime and the need for maintenance and recalibration.



Figure 4: IVR training solution in Grundfos. Retrieved from [4]

Motivation The VR training was also intended to reduce downtime and maintenance of machines while providing a safe, global, reusable tool for training. The development of the IVR training was carried out by Unity Studios, a Danish VR development company. [4].

Design Elements The VR platform includes seven different training applications, most of it targeting production workers, where training consist of completing a series of virtual versions of physical installations of machines, such as montage, measures of depth and centering, assembling construction parts, and packing a completed pump and preparing it for transportation. The IVR scenarios included a realistic representation of the machines and environment, as seen in Fig. 4, and visual cues to guide the user to perform the training tasks.

Technological Elements

- *Virtual Reality* – The VR hardware in use was the HTC Vive tracked on a room-scale. The VR environment was designed to exactly replicate the look and feel of real-world Grundfos factories. Interactions were through the Vive controllers using both their tracked positions and the controller buttons. The trainees get audio, visual, and text cues guiding them to complete the training steps.
- *Haptics* – No haptic feedback was reported.
- *User Adaptation* – No adaptation for specific users was identified.

Assessments There was no systematic validation of the IVR training. An informal contest-type assessment between IVR trained and non-IVR trained employees provided equal performance metrics, depicting the validity of IVR as a training tool.

Outcome The IVR training is in use and stakeholders testify that with the use of IVR training, they observed an increase in the motivation and decrease of the time required to train employees. In addition, the employment of IVR provided the possibility of a) increased collaboration, as several VR users could train together on the same content regardless of time and space and b) deployment of a uniform training method and scenarios on a global scale [3]. Nevertheless, some side effects like cybersickness, in a group of workers with no previous game experience, was also reported.

5 ANALYSIS AND DISCUSSION

In this section, an attempt to integrate the findings from the three cases and answer the research questions posed as part of this research is depicted.

What motivates industries to adopt IVR training?

It was evident from the cases that even big companies, leaders in their fields, as the ones investigated herein, decided to invest in IVR

training, precisely when a strong business case was made. In more detail, considerable training cost, scarce resources, downtime was parameters that made the investment in IVR training reasonable. Yet, it was also evident that the investments were dictated by an iterative process with funds released as the IVR content and features matured and tested, usually resulting in a proof-of-concept application that was then disseminated to the rest of the organisation. Other parameters reported as motivational to invest and adopt IVR training include safety, reusability, standardisation and mobility.

What technological parameters and design elements are incorporated in the IVR training?

Design and Technological Elements – We identified certain design elements during the analysis of the use cases and we categorised them as follows –

- *Hardware* – All of the use cases involved well-known commercial HMDs like the HTC Vive. Nevertheless, the interviewees expressed interest to invest in untethered solutions like the Oculus Quest due to its portability, low cost, and ease of use. No specialised haptic devices or other apparatus were utilised to augment the training experience.
- *Multi-user interactions* – Clear intentions to apply peer cooperation in some form or another was reported. Siemens-Gamesa had a multi-user experience where two users had to cooperate to operate a crane and move equipment to the wind turbine nacelle. On the other hand, in the DSB case, a pairwise training strategy with peer-feedback was applied. This may reflect a trend in industrial training, as industrial activities are essentially cooperative in nature.
- *Immersive features* – Immersive VR has certain features that make it distinct, including the angle of view, three-dimensional interaction, head, and body tracking and immersive stimuli. In two of the cases a high-fidelity HMD (HTC Vive) is used along with precise tracking of the head and the controllers which leads to immersive experiences, while in one the Oculus Quest is used which offers the same capabilities but with a slightly lesser screen resolution. Unfortunately, the direct effects of the immersion were not mapped against any other parameter and/or the resulting performance.
- *Object interaction realism* – We broadly define object interaction realism as the fidelity of the interaction with digital objects in the virtual environment. In all three cases, there is a focus on high visual fidelity with near replicas of the environment, yet it was observed that the interaction with the digital objects is not realistic, identical to the real physical objects and tools. This is attributed to the design of the experience and the utilization of the generic VR controllers, which do not represent the dimensions, weight, and other physical characteristics of the objects and tools the trainees are being trained to use. Interestingly, in the DSB and Siemens-Gamesa cases, they used haptic feedback, with the latter including haptic feedback for signifying when the trainees hit walls and in the DSB case, to signify successful completion of tasks. The use cases did not reveal any use of haptics for motor skill training.
- *Break in immersion* – One may refer to “break in immersion” as any factor that can divert the user to lose their perceived feeling of immersion in the virtual environment. The Siemens-Gamesa and Grundfos cases seem to involve immersion in the VR environment, with the aforementioned limitations of lack of object realism. However, the DSB case contains communication of the immersed in VR trainee with a fellow trainee situated outside the IVR environment (who views the IVR

scene via an iPad). One may argue that this interaction might break immersion, affecting the training performance.

How is IVR training assessed and perceived by the stakeholders?

Assessments in all the cases were oriented around either peer giving feedback to each other or experts present at the site giving feedback to the trainees or unstructured contests. There were no structured automated assessments with objective and subjective measures. Nevertheless, the perception of these proof-of-concept applications was reported as positive across the board. Even so that in all the cases intentions to increase the investment and augment the percentage of IVR training applications were reported.

6 CONCLUSION

In this research, three prominent examples of IVR training in renewable energy, transportation services, and manufacturing sectors were depicted. Such industrial examples clearly provide helpful insights into the motivation, technological and design characteristics, and perception of IVR training from a business point of view.

Our analysis depicted the value of building a business case in order to invest in IVR training, the use of commercial HMDs, the need for multi-user interactions, immersive features, object interaction realism, and immersion. Also, the lack of dedicated haptics or other apparatus and assessment methods.

Our future plans involve experimentation upon these findings in order to frame IVR training and provide clear guidelines that can augment the training performance.

ACKNOWLEDGMENTS

The researchers would like to thank all the key stakeholders interviewed for their cooperation.

REFERENCES

- [1] DSB Organization. <https://www.dsb.dk/om-dsb/virksomheden/organisationen/>. Accessed: 2021-01-24.
- [2] KANDA (XR Developer): “Turning Tool Training in Siemens-Gamesa”. <https://www.kanda.dk/turner-tool-training>. Accessed: 2021-01-24.
- [3] “Nu er VR og smartwatch hverdag på Grundfos”. <https://iu.dk/nyheder/temaer/industri-40/velkommen-til-industri-40/nu-er-vr-og-smartwatch-hverdag-paa-grundfos/>. Accessed: 2021-01-24.
- [4] Unity Studios (XR Developer): “VR Training in Grundfos”. <https://www.unity-studios.com/cases/grundfos-vr-tool/>. Accessed: 2021-01-24.
- [5] Siemens-Gamesa Turner Tool Training Social Media Update. <https://www.facebook.com/SiemensGamesa/posts/imagine-you-could-practice-your-work-in-a-virtual-offshore-wind-turbine-without-/1163625067073984/>, January 2018. Accessed: 2021-01-24.
- [6] “Grundfos Awarded Global Water And Wastewater Pumps Company Of The Year”. <https://www.grundfos.com/about-us/company>, April 2019. Accessed: 2021-01-24.
- [7] C. Anthes, R. J. García-Hernández, M. Wiedemann, and D. Kranzlmüller. State of the art of virtual reality technology. In *2016 IEEE Aerospace Conference*, pp. 1–19, 2016. doi: 10.1109/AERO.2016.7500674
- [8] P. Carlson, A. Peters, S. B. Gilbert, J. M. Vance, and A. Luse. Virtual training: Learning transfer of assembly tasks. *IEEE Transactions on Visualization and Computer Graphics*, 21(6):770–782, 2015. doi: 10.1109/TVCG.2015.2393871
- [9] A.-C. Falck, R. Örtengren, and D. Högberg. The impact of poor assembly ergonomics on product quality: A cost-benefit analysis in car manufacturing. *Human Factors and Ergonomics in Manufacturing & Service Industries*, 20(1):24–41, 2010. doi: 10.1002/hfm.20172
- [10] L. Freifeld. 2019 training industry report. *Training Magazine*, November 2019.

- [11] A. A. García, I. G. Bobadilla, G. Arroyo-Figueroa, M. Pérez-Ramírez, and J. Román. Virtual reality training system for maintenance and operation of high-voltage overhead power lines. *Virtual Reality*, 20:27–40, 2015.
- [12] A. Grabowski and J. Jankowski. Virtual reality-based pilot training for underground coal miners. *Safety Science*, 72:310 – 314, 2015. doi: 10.1016/j.ssci.2014.09.017
- [13] R. Grossman and E. Salas. The transfer of training: what really matters. *International Journal of Training and Development*, 15(2):103–120, 2011. doi: 10.1111/j.1468-2419.2011.00373.x
- [14] K. S. Gurusamy, R. Aggarwal, L. Palanivelu, and B. R. Davidson. Virtual reality training for surgical trainees in laparoscopic surgery. *Cochrane database of systematic reviews*, (1), 2009.
- [15] E. B. Hsu, Y. Li, J. D. Bayram, D. Levinson, S. Yang, and C. Monahan. State of virtual reality based disaster preparedness and response training. *PLoS currents*, 5, 2013.
- [16] J. Karim. DSB VR Training. <https://www.linkedin.com/posts/jacob-karim-54682263.dsb-vrtraining-vrededucation-ugcPost-6725030074556456960-ubb0>, November 2020. Accessed: 2021-01-24.
- [17] K. Koumaditis, F. Chinello, and S. Venckute. Design of a virtual reality and haptic setup linking arousals to training scenarios: A preliminary stage. In *2018 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*, pp. 1–2, 2018. doi: 10.1109/VR.2018.8446528
- [18] K. Koumaditis, F. Chinello, P. Mitkidis, and S. Karg. Effectiveness of virtual versus physical training: The case of assembly tasks, trainer’s verbal assistance, and task complexity. *IEEE Computer Graphics and Applications*, 40(5):41–56, 2020. doi: 10.1109/MCG.2020.3006330
- [19] A. G. Moore, X. Hu, J. C. Eubanks, A. A. Aiyaz, and R. P. McMahan. A formative evaluation methodology for vr training simulations. In *2020 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pp. 125–132, 2020. doi: 10.1109/VRW50115.2020.00027
- [20] M. Murcia-López and A. Steed. A comparison of virtual and physical training transfer of bimanual assembly tasks. *IEEE Transactions on Visualization and Computer Graphics*, 24(4):1574–1583, 2018. doi: 10.1109/TVCG.2018.2793638
- [21] T. Okuma and T. Kurata. Service field simulator: Virtual environment display system for analyzing human behavior in service fields. In *International Conference on Serviceology*, pp. 145–157. Springer, 2014.
- [22] P. H. Pucher, N. Batrick, D. Taylor, M. Chaudery, D. Cohen, and A. Darzi. Virtual-world hospital simulation for real-world disaster response: design and validation of a virtual reality simulator for mass casualty incident management. *Journal of Trauma and Acute Care Surgery*, 77(2):315–321, 2014.
- [23] J. Reason and A. Hobbs. Managing maintenance error: A practical guide. 2017.
- [24] E. Salas, S. I. Tannenbaum, K. Kraiger, and K. A. Smith-Jentsch. The science of training and development in organizations: What matters in practice. *Psychological Science in the Public Interest*, 13(2):74–101, 2012. PMID: 26173283. doi: 10.1177/1529100612436661
- [25] N. E. Seymour, A. G. Gallagher, S. A. Roman, M. K. O’Brien, V. K. Bansal, D. K. Andersen, and R. M. Satava. Virtual reality training improves operating room performance: results of a randomized, double-blinded study. *Annals of surgery*, 236(4):458, 2002.